3.2 Functional analysis: study notes

These notes condense the module. Scalars are $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. X, Y denote normed spaces, H a Hilbert space, $B(X,Y)$ the bounded linear maps, $X^* = B(X,\mathbb{F})$ the dual space. Read the texts first; use these notes to consolidate and review.

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1. Normed and Banach spaces

A *norm* satisfies $\|x\| = 0 \iff x = 0$, $\|\lambda x\| = |\lambda|\|x\|$, and the triangle inequality. A *Banach space* is a complete normed space.

Standard examples.

- ℓ^p $(1 \le p < \infty)$, ℓ^∞, and $c_0 = \{x \in \ell^\infty : x_n \to 0\}$, a closed subspace of ℓ^∞.
- $C(K)$ with the sup norm, K compact Hausdorff.
- $L^p(\mu)$, $1 \le p \le \infty$ (Riesz–Fischer, from 3.1).
- $C^k[a,b]$ with $\|f\| = \sum_{j \le k} \|f^{(j)}\|_\infty$.
- Non-example: $C[0,1]$ with the L^1 norm is not complete; its completion is $L^1[0,1]$.

Completeness criterion. X is complete iff every absolutely convergent series $\sum x_n$ (meaning $\sum \|x_n\| < \infty$) converges.

Quotients. If $M \subseteq X$ is a closed subspace, $\|x + M\| = \inf_{m \in M} \|x + m\|$ is a norm on X/M, complete if X is.

Finite dimensions. All norms on a finite-dimensional space are equivalent (compare with the ℓ^1 norm in a basis, and use compactness of its unit sphere). Consequences: finite-dimensional normed spaces are complete; finite-dimensional subspaces are closed; every linear map out of a finite-dimensional normed space is bounded.

Riesz's lemma. If $M \subsetneq X$ is a closed proper subspace and $0 < \theta < 1$, there is x with $\|x\| = 1$ and $\operatorname{dist}(x, M) \ge \theta$.

Proof. Take $x_0 \notin M$, $d = \operatorname{dist}(x_0, M) > 0$ (closedness), $m_0 \in M$ with $\|x_0 - m_0\| < d/\theta$. Put $x = (x_0 - m_0)/\|x_0 - m_0\|$. For $m \in M$, $\|x - m\| = \|x_0 - (m_0 + \|x_0 - m_0\| m)\| / \|x_0 - m_0\| \ge d/\|x_0 - m_0\| > \theta$. □

Theorem. The closed unit ball of X is compact iff $\dim X < \infty$. (If infinite-dimensional, Riesz's lemma builds unit vectors $x_1, x_2, \ldots$ with $\|x_n - x_m\| \ge \frac{1}{2}$, using $M = \operatorname{span}\{x_1, \ldots, x_{n-1}\}$.) This single fact is behind most differences between finite and infinite dimensions.

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2. Bounded linear operators

For linear $T : X \to Y$ the following are equivalent: T is continuous; continuous at 0; bounded, i.e. $\|T\| = \sup_{\|x\| \le 1} \|Tx\| < \infty$. $B(X,Y)$ with the operator norm is a normed space, complete when Y is. In particular X^* is always a Banach space. $\|ST\| \le \|S\|\|T\|$, so $B(X)$ is a Banach algebra.

Unbounded examples. Differentiation $f \mapsto f'$ from $(C^1[0,1], \|\cdot\|_\infty)$ to $C[0,1]$ ($\|x^n\|_\infty = 1$, $\|n x^{n-1}\|_\infty = n$). Every infinite-dimensional normed space has unbounded linear functionals (extend a Hamel basis with unbounded values; this uses choice).

Neumann series. If X is Banach and $\|T\| < 1$, then $I - T$ is invertible with $(I-T)^{-1} = \sum_{n \ge 0} T^n$. Hence the invertible operators form an open set, and inversion is continuous.

Duals of sequence spaces. For $1 \le p < \infty$ and $1/p + 1/q = 1$, the map $b \mapsto (a \mapsto \sum a_n b_n)$ is an isometric isomorphism $\ell^q \to (\ell^p)^*$; also $c_0^* \cong \ell^1$. But $(\ell^\infty)^* \supsetneq \ell^1$: a Banach limit (constructed with Hahn–Banach) is not given by any ℓ^1 sequence.

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3. Hilbert spaces

An inner product space satisfies **Cauchy–Schwarz** $|\langle x, y\rangle| \le \|x\|\|y\|$ and the **parallelogram law**

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

A norm comes from an inner product iff it satisfies the parallelogram law (Jordan–von Neumann); the inner product is then recovered by **polarisation**, e.g. over $\mathbb{C}$, $\langle x, y\rangle = \frac{1}{4}\sum_{k=0}^{3} i^k \|x + i^k y\|^2$ (inner product linear in the first slot). A complete inner product space is a *Hilbert space*. Examples: ℓ^2, $L^2(\mu)$, Sobolev spaces H^k.

Projection theorem. Let $C \subseteq H$ be non-empty, closed and convex. Each $x \in H$ has a unique closest point $P_C x \in C$, characterised by $\operatorname{Re}\langle x - P_C x,\, c - P_C x\rangle \le 0$ for all $c \in C$.

Proof. Let $d = \inf_{c\in C}\|x - c\|$ and $c_n \in C$ with $\|x - c_n\| \to d$. By the parallelogram law applied to $x - c_n$ and $x - c_m$,

$$\|c_n - c_m\|^2 = 2\|x-c_n\|^2 + 2\|x-c_m\|^2 - 4\left\|x - \tfrac{c_n+c_m}{2}\right\|^2 \le 2\|x-c_n\|^2 + 2\|x-c_m\|^2 - 4d^2 \to 0,$$

using convexity ($\frac{c_n+c_m}{2} \in C$). So (c_n) is Cauchy; its limit is a minimiser (closedness). Uniqueness follows from the same identity. □

Orthogonal decomposition. If M is a closed subspace, P_M is linear, $x - P_M x \in M^\perp$, and $H = M \oplus M^\perp$. Also $M^{\perp\perp} = M$; for any subspace S, $S^{\perp\perp} = \overline{S}$. A subspace is dense iff its orthogonal complement is $\{0\}$.

Riesz representation theorem. For every $\varphi \in H^*$ there is a unique $y \in H$ with $\varphi(x) = \langle x, y\rangle$ for all x, and $\|\varphi\| = \|y\|$.

Proof. If $\varphi = 0$ take $y = 0$. Otherwise $N = \ker\varphi$ is a closed proper subspace; pick $z \in N^\perp$ with $\varphi(z) = 1$. For any x, $x - \varphi(x)z \in N$, so $\langle x - \varphi(x)z, z\rangle = 0$, i.e. $\varphi(x) = \langle x, z\rangle/\|z\|^2$. Take $y = z/\|z\|^2$. □

So $H^* \cong H$ (conjugate-linearly over $\mathbb{C}$), and every Hilbert space is reflexive. This theorem gave the slick proof of Radon–Nikodym in 3.1.

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4. Orthonormal bases

An orthonormal set $\{e_\alpha\}$ satisfies **Bessel's inequality** $\sum_\alpha |\langle x, e_\alpha\rangle|^2 \le \|x\|^2$ (only countably many terms are non-zero). It is an *orthonormal basis* (maximal orthonormal set) iff any of the following equivalent conditions holds:

1. $\langle x, e_\alpha\rangle = 0$ for all α implies $x = 0$;
2. $x = \sum_\alpha \langle x, e_\alpha\rangle e_\alpha$ for all x (unconditionally convergent in norm);
3. **Parseval:** $\|x\|^2 = \sum_\alpha |\langle x, e_\alpha\rangle|^2$ for all x;
4. the linear span is dense.

Every Hilbert space has an orthonormal basis (Zorn), and all have the same cardinality (the *Hilbert dimension*). H is separable iff it has a countable orthonormal basis (Gram–Schmidt on a

countable dense set). Hence **every infinite-dimensional separable Hilbert space is isometrically isomorphic to** ℓ^2, via $x \mapsto (\langle x, e_n\rangle)_n$ (the Riesz–Fischer theorem in its original form).

Examples. $\{e^{inx}/\sqrt{2\pi}\}_{n\in\mathbb{Z}}$ in $L^2[0,2\pi]$ (completeness via Fejér or Stone–Weierstrass and density of C in L^2); normalised Legendre polynomials in $L^2[-1,1]$; Haar wavelets in $L^2[0,1]$.

Pitfall: an orthonormal basis is not a Hamel (algebraic) basis. An infinite-dimensional Banach space has no countable Hamel basis (Baire).

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5. The Hahn–Banach theorem

Analytic form (real). Let $p : X \to \mathbb{R}$ be sublinear ($p(x+y) \le p(x)+p(y)$, $p(tx) = tp(x)$ for $t \ge 0$), $M \subseteq X$ a subspace, and $f : M \to \mathbb{R}$ linear with $f \le p$ on M. Then f extends to a linear $F : X \to \mathbb{R}$ with $F \le p$ on X.

Proof sketch. One-dimensional step: to extend to $M + \mathbb{R}x_0$ we need a value $c = F(x_0)$ with $f(m) + tc \le p(m + tx_0)$ for all $m \in M$, $t \in \mathbb{R}$. Dividing by $|t|$, this is equivalent to

$$\sup_{m\in M} \left(f(m) - p(m - x_0)\right) \le c \le \inf_{m\in M} \left(p(m + x_0) - f(m)\right),$$

and the sup is at most the inf because $f(m) + f(m') = f(m+m') \le p(m+m') \le p(m - x_0) + p(m' + x_0)$. *Transfinite step:* apply Zorn's lemma to the poset of dominated extensions; a maximal one must be defined on all of X by the one-dimensional step. □

Complex form. If p is a seminorm and $|f| \le p$ on M, extend the real-linear functional $\operatorname{Re} f$ to a real-linear $G \le p$ on X, and set $F(x) = G(x) - iG(ix)$. Then F is complex-linear, extends f, and $|F| \le p$.

Normed-space corollaries.

1. Every $f \in M^*$ extends to $F \in X^*$ with $\|F\| = \|f\|$ (take $p = \|f\|\|\cdot\|$).
2. For every $x \ne 0$ there is $f \in X^*$, $\|f\| = 1$, $f(x) = \|x\|$. Hence X^* separates points and $\|x\| = \max_{\|f\|\le 1} |f(x)|$.
3. If M is a closed subspace and $x \notin M$, there is $f \in X^*$ with $f|_M = 0$ and $f(x) \ne 0$. Hence a subspace is dense iff the only functional vanishing on it is 0.
4. If X^* is separable, so is X.

Geometric form (separation). If $A, B \subseteq X$ are disjoint convex sets with A open, there are $f \in X^*$ and $\gamma \in \mathbb{R}$ with $\operatorname{Re} f(a) < \gamma \le \operatorname{Re} f(b)$ for all $a \in A$, $b \in B$. If A is closed and B is compact (both convex, disjoint), they can be strictly separated. The proof applies the analytic form to the Minkowski functional of an open convex set.

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6. Baire category and its three consequences

Baire category theorem. In a complete metric space, a countable intersection of dense open sets is dense. Equivalently, a complete metric space is not a countable union of nowhere dense sets.

Uniform boundedness principle (Banach–Steinhaus). Let X be Banach and $\mathcal{F} \subseteq B(X,Y)$. If $\sup_{T\in\mathcal{F}} \|Tx\| < \infty$ for each $x \in X$, then $\sup_{T\in\mathcal{F}} \|T\| < \infty$.

Proof. The sets $E_n = \{x : \sup_T \|Tx\| \le n\}$ are closed and cover X; by Baire, some E_n contains a ball $B(x_0, r)$. For $\|z\| < r$, $\|Tz\| \le \|T(x_0+z)\| + \|Tx_0\| \le 2n$, so $\|T\| \le 2n/r$ for all $T \in \mathcal{F}$. □

Consequences: weakly convergent sequences are bounded; a pointwise limit of bounded operators on a Banach space is bounded; there is a continuous function whose Fourier series diverges at a point (the Dirichlet kernels have L^1 norms $\to \infty$).

Open mapping theorem. If X, Y are Banach and $T \in B(X,Y)$ is surjective, then T is open (maps open sets to open sets).

Proof sketch. $Y = \bigcup_n \overline{T(B_X(0,n))}$, so by Baire some $\overline{T(B_X(0,n))}$ has interior, and by symmetry and convexity $\overline{T(B_X(0,1))} \supseteq B_Y(0,\delta)$. A successive-approximation argument using completeness of X removes the closure: $T(B_X(0,2)) \supseteq B_Y(0,\delta)$. □

Bounded inverse theorem. A bijective bounded operator between Banach spaces has bounded inverse. Corollary: if a vector space is complete under two norms with $\|\cdot\|_1 \le C\|\cdot\|_2$, the norms are equivalent.

Closed graph theorem. If X, Y are Banach and $T : X \to Y$ is linear with closed graph $\{(x, Tx)\} \subseteq X \times Y$, then T is bounded. (Apply the bounded inverse theorem to the projection from the graph onto X.) In practice: to show T bounded, it suffices to show that $x_n \to x$ and $Tx_n \to y$ imply $y = Tx$.

Pitfall: all three theorems need completeness. On non-complete spaces each can fail.

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7. Duals and reflexivity

The canonical map $J : X \to X^{**}$, $J(x)(f) = f(x)$, is an isometry (Hahn–Banach, corollary 2). X is *reflexive* if J is onto.

- Hilbert spaces are reflexive (Riesz representation twice).
- L^p for $1 < p < \infty$ is reflexive, since $(L^p)^* = L^q$ and $(L^q)^* = L^p$ through the same pairing.
- c_0, ℓ^1, ℓ^∞, $L^1[0,1]$, $L^\infty[0,1]$ and $C[0,1]$ are not reflexive.
- Closed subspaces of reflexive spaces are reflexive. X is reflexive iff X^* is.
- If X is separable and reflexive then X^* is separable; so a separable space with non-separable dual (like ℓ^1, with dual ℓ^∞) is not reflexive.

Dual of L^p. For σ-finite μ and $1 \le p < \infty$, $(L^p)^* \cong L^q$. Sketch for finite μ: given $\varphi \in (L^p)^*$, $\nu(E) = \varphi(\chi_E)$ is a (complex) measure, absolutely continuous with respect to μ; its Radon–Nikodym derivative g represents φ on simple functions; then show $g \in L^q$ by testing against truncations of $|g|^{q-1}\operatorname{sgn}\bar{g}$.

Adjoint (Banach-space version). For $T \in B(X,Y)$, $T' : Y^* \to X^*$, $T'f = f \circ T$, with $\|T'\| = \|T\|$.

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8. Weak and weak* topologies

The **weak topology** on X is the coarsest making every $f \in X^*$ continuous: $x_n \rightharpoonup x$ iff $f(x_n) \to f(x)$ for all $f \in X^*$. The **weak* topology** on X^* is the coarsest making every evaluation $f \mapsto f(x)$ continuous: $f_n \overset{*}{\rightharpoonup} f$ iff $f_n(x) \to f(x)$ for all x.

Facts:

- In infinite dimensions, the weak topology is strictly coarser than the norm topology; every weak neighbourhood of 0 contains a non-trivial subspace, so is unbounded. The weak topology is not metrisable (on infinite-dimensional X).
- Weakly convergent sequences are bounded (UBP) and $\|x\| \le \liminf \|x_n\|$.

- In Hilbert space, $x_n \to x$ in norm iff $x_n \rightharpoonup x$ and $\|x_n\| \to \|x\|$. Any orthonormal sequence satisfies $e_n \rightharpoonup 0$ (Bessel), but $\|e_n\| = 1$.
- **Mazur's theorem:** a convex set is weakly closed iff it is norm closed (Hahn–Banach separation). Corollary: if $x_n \rightharpoonup x$, some sequence of convex combinations of the x_n converges to x in norm.
- **Schur's theorem:** in ℓ^1, weakly convergent sequences converge in norm (even though the topologies differ!).

Banach–Alaoglu theorem. The closed unit ball of X^* is weak* compact.

Proof. Map B_{X^*} into $P = \prod_{x \in X} \{z \in \mathbb{F} : |z| \le \|x\|\}$ by $f \mapsto (f(x))_x$. This is a homeomorphism onto its image when B_{X^*} has the weak* topology and P the product topology. P is compact by Tychonoff. The image consists of the points of P satisfying the linearity relations $z_{x+y} = z_x + z_y$, $z_{\lambda x} = \lambda z_x$, each a closed condition, so the image is closed in P, hence compact. □

Sequential versions.

- If X is separable, the weak* topology on B_{X^*} is metrisable, so every bounded sequence in X^* has a weak* convergent subsequence. (Direct diagonal argument along a countable dense set.)
- If X is reflexive, every bounded sequence in X has a weakly convergent subsequence (Eberlein–Šmulian; for separable reflexive X apply the previous point to X^*, which is separable). In particular this holds in every Hilbert space and in L^p, $1 < p < \infty$.

Why this matters. The direct method in the calculus of variations: a bounded minimising sequence has a weakly convergent subsequence, and a convex, lower semicontinuous functional is weakly lower semicontinuous (Mazur), so the weak limit is a minimiser. This is the existence proof for weak solutions of many PDEs.

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9. Adjoints and spectrum

Hilbert adjoint. For $T \in B(H_1, H_2)$, $T^* \in B(H_2, H_1)$ is defined by $\langle Tx, y\rangle = \langle x, T^* y\rangle$ (exists by Riesz). Properties: $T^{**} = T$, $\|T^*\| = \|T\|$, and the **C*-identity** $\|T^*T\| = \|T\|^2$. Also

$$\ker T^* = (\operatorname{ran} T)^\perp, \qquad \overline{\operatorname{ran} T} = (\ker T^*)^\perp.$$

T is *self-adjoint* if $T = T^*$; *normal* if $TT^* = T^*T$; *unitary* if $T^*T = TT^* = I$; an *orthogonal projection* if $T = T^2 = T^*$.

For self-adjoint T: $\langle Tx, x\rangle \in \mathbb{R}$ and

$$\|T\| = \sup_{\|x\|=1} |\langle Tx, x\rangle|.$$

(Proof of $\le$: with M the right side, polarisation gives $4\operatorname{Re}\langle Tx, y\rangle = \langle T(x+y), x+y\rangle - \langle T(x-y), x-y\rangle \le M(\|x+y\|^2 + \|x-y\|^2) = 2M(\|x\|^2 + \|y\|^2)$; take $\|x\| = 1$, $y = Tx/\|Tx\|$.)

Spectrum. For $T \in B(X)$, X a complex Banach space, $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I$ not invertible in $B(X)\}$. (By the bounded inverse theorem, "invertible" just means bijective.)

- $\sigma(T)$ is compact and contained in $\{|\lambda| \le \|T\|\}$ (Neumann series), and **non-empty** (otherwise $\lambda \mapsto f((T-\lambda)^{-1})$ is a bounded entire function tending to 0, so zero for every $f \in B(X)^*$ — Liouville plus Hahn–Banach).
- **Spectral radius formula (Gelfand):** $r(T) = \max_{\lambda \in \sigma(T)} |\lambda| = \lim_n \|T^n\|^{1/n}$.
- $\sigma(T)$ splits into the **point spectrum** (eigenvalues: $T - \lambda$ not injective), **continuous spectrum** (injective, dense non-closed range) and **residual spectrum** (injective, range not dense).

- If T is self-adjoint, $\sigma(T) \subseteq [m, M] \subseteq \mathbb{R}$ with $m = \inf_{\|x\|=1}\langle Tx, x\rangle$, $M = \sup_{\|x\|=1}\langle Tx, x\rangle$, and $m, M \in \sigma(T)$. Eigenvectors for distinct eigenvalues are orthogonal. Normal operators have $r(T) = \|T\|$.

Key examples on ℓ^2.

- Right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$: an isometry, not onto; no eigenvalues; $\sigma(S) = \overline{\mathbb{D}}$.
- Left shift S^*: every $|\lambda| < 1$ is an eigenvalue (eigenvector $(1, \lambda, \lambda^2, \dots)$); $\sigma(S^*) = \overline{\mathbb{D}}$.
- Multiplication $M_\varphi f = \varphi f$ on $L^2[0,1]$, φ continuous: $\sigma(M_\varphi) = \varphi([0,1])$; eigenvalues only where φ is constant on a set of positive measure. For $\varphi(x) = x$ there are no eigenvalues and $\sigma = [0,1]$. This is the model for the spectral theorem for general self-adjoint operators.

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10. Compact operators

$T \in B(X, Y)$ is **compact** if $T(B_X)$ is relatively compact; equivalently (for Banach Y) every bounded sequence (x_n) has a subsequence with (Tx_{n_k}) convergent.

- Finite-rank bounded operators are compact. Compact operators form a closed two-sided ideal $K(X)$ in $B(X)$: closed under sums, under composition with bounded operators on either side, and under norm limits (diagonal argument / total boundedness).
- **In Hilbert space,** compact = norm limit of finite-rank operators (approximate by $P_n T$ with P_n the projection onto the span of the first n basis vectors). This approximation property fails in some Banach spaces (Enflo).
- T compact $\iff$ T^* compact (Schauder).
- In Hilbert space, T is compact iff $x_n \rightharpoonup 0 \Rightarrow Tx_n \to 0$ in norm. Compact operators turn weak convergence into norm convergence.
- The identity on an infinite-dimensional space is not compact (Riesz). Hence a compact operator on an infinite-dimensional space is never invertible and $0 \in \sigma(T)$.

Examples.

- Diagonal operator $De_n = \lambda_n e_n$ on ℓ^2 is compact iff $\lambda_n \to 0$.
- **Hilbert–Schmidt integral operators:** if $k \in L^2([0,1]^2)$, then $(Kf)(x) = \int_0^1 k(x,y) f(y)\, dy$ is compact on $L^2[0,1]$ with $\|K\| \le \|k\|_{L^2}$. (Expand k in the orthonormal basis $e_i(x)\overline{e_j(y)}$ and truncate.) K^* has kernel $\overline{k(y,x)}$; K is self-adjoint iff $k(x,y) = \overline{k(y,x)}$.
- **Volterra operator** $(Vf)(x) = \int_0^x f$: compact, with no eigenvalues and $\sigma(V) = \{0\}$ ($\|V^n\| \le 1/n!$).

Riesz–Schauder theory. If $K \in B(X)$ is compact and $\lambda \neq 0$: $\ker(K - \lambda)$ is finite-dimensional, $\operatorname{ran}(K - \lambda)$ is closed, and every non-zero $\lambda \in \sigma(K)$ is an eigenvalue. The non-zero spectrum is at most countable, with 0 the only possible accumulation point.

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11. The spectral theorem for compact self-adjoint operators

Theorem. Let T be a compact self-adjoint operator on a Hilbert space H. Then there is an orthonormal sequence (e_n) (finite or countable) of eigenvectors with non-zero real eigenvalues λ_n, $|\lambda_1| \ge |\lambda_2| \ge \cdots$, $\lambda_n \to 0$ if infinitely many, such that

$$Tx = \sum_n \lambda_n \langle x, e_n\rangle e_n \qquad \text{for all } x \in H,$$

and $H = \ker T \oplus \overline{\operatorname{span}}\{e_n\}$. Adding an orthonormal basis of $\ker T$ gives an orthonormal basis of H consisting of eigenvectors.

Proof.

1. **Key lemma:** $\|T\|$ or $-\|T\|$ is an eigenvalue. Assume $T \neq 0$. By §9, pick unit x_n with $\langle Tx_n, x_n\rangle \to \lambda$ where $|\lambda| = \|T\|$. Then

$$\|Tx_n - \lambda x_n\|^2 = \|Tx_n\|^2 - 2\lambda\langle Tx_n, x_n\rangle + \lambda^2 \leq 2\lambda^2 - 2\lambda\langle Tx_n, x_n\rangle \to 0.$$

   By compactness, a subsequence $Tx_{n_k} \to y$. Then $\lambda x_{n_k} \to y$, so $\|y\| = |\lambda| \neq 0$ and, by continuity, $Ty = \lim T(\lambda x_{n_k}) = \lambda y$.
2. **Induction:** let e_1 be a unit eigenvector for $\lambda_1 = \pm\|T\|$. $H_1 = \{e_1\}^\perp$ is T-invariant (self-adjointness), and $T|_{H_1}$ is compact self-adjoint. Repeat, getting $|\lambda_1| \geq |\lambda_2| \geq \cdots$. Stop if $T|_{H_n} = 0$.
3. $\lambda_n \to 0$**:** otherwise $|\lambda_n| \geq \varepsilon$ and $\|Te_n - Te_m\|^2 = \lambda_n^2 + \lambda_m^2 \geq 2\varepsilon^2$, so (Te_n) has no convergent subsequence, contradicting compactness.
4. **Expansion:** let $M = \overline{\operatorname{span}}\{e_n\}$. T restricted to $M^\perp$ has norm at most $|\lambda_n|$ for every n (since $M^\perp \subseteq H_n$), so it is 0: $M^\perp \subseteq \ker T$. The expansion follows by writing $x = P_M x + (x - P_M x)$. □

Variational characterisation. $\lambda_1^+ = \max_{\|x\|=1}\langle Tx, x\rangle$, and the Courant–Fischer min-max formula gives the others. This is the infinite-dimensional Rayleigh quotient.

Functional calculus. For bounded f on $\sigma(T)$, define $f(T)x = \sum f(\lambda_n)\langle x, e_n\rangle e_n + f(0)P_{\ker T}x$. For example, a positive compact operator has a unique positive square root.

Application (Sturm–Liouville / Fourier series). The operator $-u'' = f$, $u(0) = u(1) = 0$ is inverted by the integral operator with Green's kernel $G(x, y) = \min(x, y) - xy$, which is Hilbert–Schmidt and self-adjoint. The spectral theorem for G yields the orthonormal basis $\sqrt{2}\sin(n\pi x)$ of $L^2[0, 1]$: that is, completeness of the Fourier sine series.

Beyond compact operators. Every bounded self-adjoint (or normal) operator is unitarily equivalent to a multiplication operator M_φ on some $L^2(\mu)$; equivalently there is a projection-valued measure E with $T = \int \lambda\, dE(\lambda)$. This is the full spectral theorem (an elective topic).

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12. The Fredholm alternative

Theorem. Let K be compact on a Hilbert space H (the statement also holds on Banach spaces with K' in place of K^*) and $\lambda \in \mathbb{C} \setminus \{0\}$. Then:

1. $\ker(K - \lambda)$ and $\ker(K^* - \bar\lambda)$ are finite-dimensional and have the same dimension;
2. $\operatorname{ran}(K - \lambda)$ is closed and equals $\ker(K^* - \bar\lambda)^\perp$;
3. **Alternative:** either $(K - \lambda)u = f$ has a unique solution for every $f \in H$, or the homogeneous equation $(K - \lambda)u = 0$ has a non-zero solution. In the second case, $(K - \lambda)u = f$ is solvable iff $f \perp \ker(K^* - \bar\lambda)$.

Equivalently: $I - K$ is injective iff surjective, and it is a Fredholm operator of index 0.

Proof ideas. (1) The unit ball of $\ker(K - \lambda)$ is mapped by K to λ times itself, so it is compact, so the kernel is finite-dimensional (Riesz). (2) If $(K - \lambda)u_n \to f$, choose $u_n \perp \ker(K - \lambda)$. If (u_n) were unbounded, normalise and use compactness to produce a unit vector in the kernel orthogonal to the kernel. So (u_n) is bounded, a subsequence of Ku_n converges, hence $u_n = \lambda^{-1}(Ku_n - (K - \lambda)u_n)$ converges to some u with $(K - \lambda)u = f$. The closed range equals $\ker(K^* - \bar\lambda)^\perp$ by §9. (3) Injective ⇒ surjective: if not, $H_n = (K - \lambda)^n H$ is a strictly decreasing

chain of closed subspaces; Riesz's lemma gives unit $x_n \in H_n \ominus H_{n+1}$, and $\|Kx_n - Kx_m\| \geq |\lambda|$ for $n < m$, contradicting compactness. Equality of dimensions in (1) follows with a finite-rank perturbation argument. □

Application: integral equations. For a Hilbert–Schmidt kernel k, the Fredholm equation of the second kind

$$u(x) - \int_0^1 k(x,y)u(y)\,dy = f(x)$$

has a unique solution in L^2 for every f provided the homogeneous equation has only the zero solution. This is the historical origin of the whole subject (Fredholm 1903, Hilbert 1904–1910).

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13. Common pitfalls

1. **Assuming linear maps are continuous.** They are in finite dimensions only. Always check boundedness.
2. **Using Baire-category theorems without completeness.** UBP, open mapping and closed graph all fail on incomplete spaces.
3. **Confusing weak, weak* and norm convergence.** In $\ell^1 = c_0^*$, the sequence e_n converges weak* to 0 but not weakly (test against $(1,1,1,\dots) \in \ell^\infty$).
4. **Thinking Banach–Alaoglu gives sequential compactness in general.** It does for separable X. For $X = \ell^\infty$, the sequence of coordinate functionals in $(\ell^\infty)^*$ has no weak* convergent subsequence.
5. **Assuming every bounded operator has an eigenvalue** (the shift does not) or that a self-adjoint operator has an orthonormal basis of eigenvectors (multiplication by x does not). Both are true for compact self-adjoint operators.
6. **Forgetting conjugate-linearity** of the Riesz map $H \to H^*$ over $\mathbb{C}$.
7. **"Orthonormal basis" vs "Hamel basis."** In infinite dimensions they are different notions.
8. **Applying the spectral theorem to compact operators that are not self-adjoint (or normal).** The Volterra operator is compact with spectrum $\{0\}$ and no eigenvectors at all.

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14. Connections forward

- **PDE:** Lax–Milgram (a coercive bounded bilinear form on a Hilbert space is represented by an invertible operator; a direct consequence of Riesz representation) gives weak solutions of elliptic equations on Sobolev spaces. Rellich's compact embedding $H_0^1 \hookrightarrow L^2$ plus the spectral theorem gives eigenfunction expansions for the Laplacian on bounded domains.
- **Operator algebras and quantum mechanics:** $B(H)$ is a C*-algebra. Observables are self-adjoint (often unbounded) operators; the full spectral theorem and Stone's theorem on one-parameter unitary groups are the next steps.
- **Harmonic analysis:** Fourier series and transforms are unitary maps (L^2 theory, Plancherel); Riesz–Thorin interpolation gives L^p bounds.
- **Probability:** L^2 martingales, conditional expectation as an orthogonal projection, weak convergence of measures (weak* convergence in $C(K)^*$).
- **Algebra and topology:** Fredholm index is a homotopy invariant, which leads to K-theory and the Atiyah–Singer index theorem, connecting to 3.4 and 3.5. The representation theory of compact groups (Peter–Weyl) generalises the finite-group theory of 3.3 using compact self-adjoint operators on $L^2(G)$.